Ex. 8. ***Simplify:*** *(a)* $108 \div 36 \text{ of } \frac{1}{4} + \frac{2}{5} \times 3\frac{1}{4}$

(b) $\frac{2}{3} \times \frac{5}{6} + \frac{4}{9} - \frac{3}{4} + \frac{2}{9} \times \frac{5}{9} \div \frac{2}{9}$ **(P.C.S., 2009)**

Sol. (*a*) Given expression $= 108 \div 9 + \frac{2}{5} \times \frac{13}{4} = \frac{108}{9} + \frac{13}{10} = \left(12 + \frac{13}{10}\right) = \frac{133}{10} = 13\frac{3}{10}$.

(*b*) Given expression $= \frac{2}{3} \times \frac{5}{6} + \frac{4}{9} - \frac{3}{4} + \frac{2}{9} \times \frac{5}{9} \times \frac{9}{2}$

$$= \frac{5}{9} + \frac{4}{9} - \frac{3}{4} + \frac{5}{9} = \frac{14}{9} - \frac{3}{4} = \frac{56-27}{36} = \frac{29}{36}.$$

Ex. 9. ***What value will replace the question mark in the following equation?***

$$4\frac{1}{2} + 3\frac{1}{6} + ? + 2\frac{1}{3} = 13\frac{2}{5}.$$

Sol. Let $\frac{9}{2} + \frac{19}{6} + x + \frac{7}{3} = \frac{67}{5}$.

Then, $x = \frac{67}{5} - \left(\frac{9}{2} + \frac{19}{6} + \frac{7}{3}\right) \Leftrightarrow x = \frac{67}{5} - \left(\frac{27+19+14}{6}\right) = \left(\frac{67}{5} - \frac{60}{6}\right)$

$$\Leftrightarrow x = \left(\frac{67}{5} - 10\right) = \frac{17}{5} = 3\frac{2}{5}.$$

Hence, missing fraction $= 3\frac{2}{5}$.

Ex. 10. ***Simplify:*** $\left[3\frac{1}{4} \div \left\{1\frac{1}{4} - \frac{1}{2}\left(2\frac{1}{2} - \overline{\frac{1}{4} - \frac{1}{6}}\right)\right\}\right]$.

Sol. Given exp. $= \left[\frac{13}{4} \div \left\{\frac{5}{4} - \frac{1}{2}\left(\frac{5}{2} - \frac{3-2}{12}\right)\right\}\right] = \left[\frac{13}{4} \div \left\{\frac{5}{4} - \frac{1}{2}\left(\frac{5}{2} - \frac{1}{12}\right)\right\}\right]$

$$= \left[\frac{13}{4} \div \left\{\frac{5}{4} - \frac{1}{2}\left(\frac{30-1}{12}\right)\right\}\right] = \left[\frac{13}{4} \div \left\{\frac{5}{4} - \frac{29}{24}\right\}\right]$$

$$= \left[\frac{13}{4} \div \left\{\frac{30-29}{24}\right\}\right] = \left[\frac{13}{4} \div \frac{1}{24}\right] = \left[\frac{13}{4} \times 24\right] = 78.$$

Ex. 11. ***Simplify:*** $\dfrac{\frac{7}{2} \div \frac{5}{2} \times \frac{3}{2}}{\frac{7}{2} \div \frac{5}{2} \text{ of } \frac{3}{2}} \div 5.25$

Sol. Given exp. $= \dfrac{\frac{7}{2} \times \frac{2}{5} \times \frac{3}{2}}{\frac{7}{2} \div \frac{15}{4}} \div 5.25 = \dfrac{\frac{21}{10}}{\frac{7}{2} \times \frac{4}{15}} \div \frac{525}{100} = \frac{21}{10} \times \frac{15}{14} \times \frac{100}{525} = \frac{6}{14} = \frac{3}{7}$.

Ex. 12. ***Simplify:*** $b - [b - (a + b) - \{b - (b - \overline{a - b})\} + 2a]$.

Sol. Given exp. $= b - [b - (a + b) - \{b - (b - a + b)\} + 2a]$

$$= b - [b - a - b - \{b - (2b - a)\} + 2a]$$

$$= b - [- a - \{b - 2b + a\} + 2a]$$

$$= b - [- a - \{- b + a\} + 2a]$$

$$= b - [- a + b - a + 2a] = b - b = 0.$$